Linear and Logistic Regression

Test Question Bank with Solutions

Conventions: Residuals are defined as prediction minus observed value.

For logistic regression, use threshold 0.5 and natural logarithms unless stated otherwise.

$$r^{(i)} = h_\theta(\mathbf{x}^{(i)}) - y^{(i)}.$$

For logistic regression, unless otherwise stated, use a classification threshold of 0.5 and natural logarithms.

Part I: Question Bank

A. Linear Regression

L1. A one-feature linear model is

$$h_\theta(x) = \theta_0 + \theta_1 x.$$

If $\theta_0 = 1.5$, $\theta_1 = 0.8$, and $x = 4$, compute the prediction.

L2. A two-feature linear model is

$$h_\theta(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2.$$

If $\theta = (-2, 1.5, 0.5)$ and $(x_1, x_2) = (2, 6)$, compute the prediction.

L3. Consider the observations

$$(1, 2), \qquad (2, 2.5), \qquad (3, 4)$$

and the model $h_A(x) = 1 + x$. Compute all three predictions, all three residuals, and the least-squares cost

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2.$$

L4. For the same observations, consider $h_B(x) = 1.5 + 0.7x$. Compute its least-squares cost and determine whether h_A or h_B has the smaller cost.

L5. For a single observation, let $x = 2$, $y = 5$, $\theta_0 = 1$, and $\theta_1 = 0.5$. Compute the prediction, the residual, and the single-observation cost

$$J^{(i)}(\theta) = \frac{1}{2} \left(h_\theta(x) - y \right)^2.$$

L6. Using the same values as in L5, compute

$$\frac{\partial J^{(i)}}{\partial \theta_0} \qquad \text{and} \qquad \frac{\partial J^{(i)}}{\partial \theta_1}.$$

L7. Using the gradient from L6 and learning rate $\alpha = 0.1$, perform one gradient-descent update starting from $(\theta_0, \theta_1) = (1, 0.5)$.

L8. Use the dataset

$$(-1, 0), \qquad (0, 1), \qquad (1, 2).$$

For the model $h(x) = \theta_0 + \theta_1 x$, compute the *batch* gradient at $(\theta_0, \theta_1) = (0, 0)$.

L9. Using your answer from L8 and $\alpha = 0.25$, perform one batch gradient-descent update.

L10. Starting from the parameter values found in L9, compute the next batch gradient and then perform the second batch update using $\alpha = 0.25$.

L11. For the dataset in L8, compute the full-data least-squares cost at $(\theta_0, \theta_1) = (0.75, 0.50)$.

L12. Starting from $(\theta_0, \theta_1) = (0, 0)$, use stochastic gradient descent with $\alpha = 0.25$ and process the observations in this order:

$$(0, 1), \qquad (1, 2), \qquad (-1, 0).$$

Show the parameter vector after each of the three updates.

L13. In one or two sentences, state the computational difference between batch gradient descent and stochastic gradient descent. Your answer must say when the parameters are updated.

L14. What can happen when the learning rate α is (a) too small and (b) too large?

L15. You are *not* asked to derive the normal equations. Use the final formula

$$\theta = (X^T X)^{-1} X^T \mathbf{y}.$$

Suppose

$$(X^T X)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}, \qquad X^T \mathbf{y} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Compute θ.

L16. A fitted model is $h(x) = 1 + x$. Predict the response at $x = 2.5$. If the actual response is 3.8, compute the residual using $r = h(x) - y$.

L17. Matrix method for least squares. Consider the training observations

x	0	1	2
y	1	2	2

Fit the model $h_{\vec{\theta}}(x) = \theta_0 + \theta_1 x$ using the matrix method. (a) Form the design matrix X with an intercept column and the target vector $\vec{y}$. (b) Compute $X^T X$ and $X^T \vec{y}$. (c) Find $(X^T X)^{-1}$. (d) Use $\vec{\theta} = (X^T X)^{-1} X^T \vec{y}$ to find θ_0 and θ_1. (e) State the fitted line and predict y when $x = 3$.

L18. Why do we square the residuals in the least-squares cost rather than simply add the residuals? Explain two reasons.

L19. Describe the roles of the training, validation, and test sets. Why should the test set not be used repeatedly to select a model?

L20. Two fitted linear models have the same training error, but one has much larger test error. What does this suggest? Explain why training error alone cannot establish predictive performance.

L21. Explain the difference between the parameter θ_1 and a prediction $h_{\vec{\theta}}(x)$ in a one-feature linear model. What does a negative θ_1 mean?

L22. A student says, "Gradient descent finds the answer by trying all possible values of $\vec{\theta}$." Explain why this statement is incorrect and describe how the gradient is used.

L23. A student fits $h_\theta(x) = \theta_0 + \theta_1 x$ to the observations $(1, 2)$ and $(2, 4)$. At $\theta_0 = 0$ and $\theta_1 = 1$, the student claims that the least-squares cost is zero because the residuals are "small." Calculate the residuals and the cost $J = \frac{1}{2}\sum_i (h_\theta(x^{(i)}) - y^{(i)})^2$. Is the student's conclusion correct?

L24. A student applies batch gradient descent to a least-squares problem. Starting from $\vec{\theta} = (0, 0)$, the cost changes from $J = 12$ to $J = 20$ and then $J = 35$ after two steps. The student concludes that the gradient formula must be wrong. Is that conclusion justified? Explain one other possible cause and what could be changed to investigate it.

L25. For one training observation $(x, y) = (2, 5)$, let $h_\theta(x) = \theta_0 + \theta_1 x$ and start from $(\theta_0, \theta_1) = (0, 0)$. With learning rate $\alpha = 0.1$, a student updates θ_0 first and then uses the new value of θ_0 when computing the update to θ_1. Explain the error. Compute the correct simultaneous gradient-descent update for $J^{(i)} = \frac{1}{2}(h_\theta(x) - y)^2$.

L26. A student fits two models and reports the following mean squared errors:

	Training	Validation	Test
Model A	1.0	2.4	2.6
Model B	0.3	4.8	5.1

Which model would have been preferable based on validation performance? What do the training and validation errors suggest about Model B? Explain why the final decision should not be made by repeatedly choosing whichever model has the better test score.

L27. For a one-feature model $h_\theta(x) = \theta_0 + \theta_1 x$, a student states: "If the gradient of J with respect to θ_1 is positive, gradient descent must increase θ_1." Evaluate the claim using the update rule. If $\theta_1 = 1$, $\partial J / \partial \theta_1 = 3$, and $\alpha = 0.2$, calculate the new value.

B. Logistic Regression

G1. Explain why the unrestricted output of an ordinary linear model is not suitable by itself as a probability for a binary outcome.

G2. For one feature, the logistic-regression linear score is

$$z = \theta_0 + \theta_1 x.$$

If $\theta_0 = -1.2$ and $\theta_1 = 0.8$, compute z at (a) $x = 3$ and (b) $x = 0$.

G3. The sigmoid function is

$$g(z) = \frac{1}{1 + e^{-z}}.$$

Using the two scores from G2, compute the two probabilities to three decimal places.

G4. Suppose a logistic model gives $h_\theta(\mathbf{x}) = 0.80$. State $P(y = 1 \mid \mathbf{x})$, state $P(y = 0 \mid \mathbf{x})$, and give the predicted class at threshold 0.5.

G5. Let $z = -2 + 0.5x$. Find the one-feature decision boundary for a threshold of 0.5.

G6. Let

$$z = -3 + x_1 + 0.5x_2.$$

Write the two-feature decision-boundary equation. Then classify the point $(x_1, x_2) = (2, 4)$ using threshold 0.5. Also compute its predicted probability to three decimal places.

G7. A predictor reports class-1 probabilities

$$0.90, \quad 0.20, \quad 0.70, \quad 0.40$$

for observations whose true labels are

$$1, \quad 0, \quad 1, \quad 0.$$

For each observation, write the probability assigned to the outcome that actually occurred. Then compute the likelihood.

G8. A second predictor reports class-1 probabilities

$$0.60, \quad 0.40, \quad 0.55, \quad 0.45$$

for the same true labels as G7. Compute its likelihood and determine which predictor gives the larger likelihood.

G9. A logistic model gives probabilities

$$0.20, \quad 0.70, \quad 0.90$$

for observations with labels

$$0, \quad 1, \quad 1.$$

Compute the likelihood and then the log-likelihood.

G10. Starting from

$$g(z) = \frac{1}{1 + e^{-z}},$$

show that

$$g'(z) = g(z)(1 - g(z)).$$

G11. For a single observation with $y = 1$ and predicted class-1 probability $h = 0.80$, compute the negative log-likelihood loss.

G12. For a single observation with $y = 0$ and predicted class-1 probability $h = 0.80$, compute the negative log-likelihood loss.

G13. Verify G12 using the single binary cross-entropy formula

$$J^{(i)} = -\left[y \log h + (1 - y) \log(1 - h)\right].$$

Use $y = 0$ and $h = 0.80$.

G14. For one training observation, suppose

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad y = 1, \qquad h_\theta(\mathbf{x}) = 0.60.$$

The current parameter vector is

$$\theta = \begin{bmatrix} -0.5 \\ 0.4 \end{bmatrix}$$

and $\alpha = 0.1$. Compute the stochastic gradient-ascent vector

$$(y - h)\mathbf{x}$$

and then perform one gradient-ascent update.

G15. Use the dataset

x	-2	-1	0	1	2	3
y	0	0	1	0	1	1

with an intercept. Start from $\theta = (0, 0)^T$, so all initial probabilities are 0.5. Compute the *average* log-likelihood gradient

$$\frac{1}{m} \sum_{i=1}^{m} (y^{(i)} - h^{(i)}) \mathbf{x}^{(i)}.$$

Then use $\alpha = 0.8$ to perform one batch gradient-ascent update.

G16. Using the parameter vector from G15, compute the score z, the class-1 probability, and the predicted class for $x = 2$.

G17. For negative log-likelihood, the single-observation gradient is

$$(h - y)\mathbf{x}.$$

Suppose $y = 0$, $h = 0.70$, $\mathbf{x} = (1, 3)^T$, current $\theta = (0.2, -0.1)^T$, and $\alpha = 0.05$. Compute the gradient and perform one gradient-descent update.

G18. Explain algebraically why maximizing log-likelihood by gradient ascent and minimizing negative log-likelihood by gradient descent produce the same parameter-update direction.

G19. Why is ordinary linear regression generally unsuitable for predicting a binary outcome coded as 0 and 1? Explain what the sigmoid changes, and distinguish the probability from the final predicted class.

G20. A logistic model produces $h_{\vec{\theta}}(\vec{x}) = 0.74$. Explain what this means, state the probability of class 0, and explain why 0.74 is not itself a class label.

G21. Why does the decision boundary of a logistic-regression model with two features form a line even though the sigmoid curve is nonlinear?

G22. Two binary classifiers predict the correct labels for every example. Classifier A gives probabilities 0.90 and 0.10 for examples whose actual labels are 1 and 0; classifier B gives 0.55 and 0.45 for those same examples. Explain why their accuracy is the same but their likelihoods differ. Calculate both likelihoods.

G23. Explain why a confidently incorrect logistic prediction produces a larger negative-log-likelihood loss than a mildly incorrect prediction. Give a numerical example for a case with true label $y = 1$.

G24. What is the effect of changing the classification threshold from 0.5 to 0.7? Does it change the probabilities calculated by the already-fitted logistic model? Explain.

G25. A logistic-regression model has linear score $z = -2$. A student writes: "The predicted probability is -2, so the predicted class is 0." Identify the error, compute the actual probability $P(y = 1 \mid \vec{x})$ to three decimal places, and give the predicted class with threshold 0.5.

G26. Two logistic models produce class-1 probabilities 0.51 and 0.99 for an observation whose true label is $y = 1$. At threshold 0.5, do they have different classification accuracy on this observation? Compute the negative log-likelihood loss of each and explain the difference.

G27. A student claims that the likelihood contribution from an observation with true label $y = 0$ equals the model's output $h_\theta(\vec{x})$. If $h_\theta(\vec{x}) = 0.15$, calculate the correct likelihood contribution and the corresponding negative log-likelihood loss. Explain the correction.

G28. Consider $h_\theta(x) = g(-1 + 0.5x)$, where $g(z) = 1/(1 + e^{-z})$. A student claims the 0.5 decision boundary is at $x = 0$ because $g(0) = 0.5$. Find the actual boundary and explain why it occurs there.

G29. For one logistic training observation with $y = 1$, $\vec{x} = (1, 2)^T$, and current predicted probability $h = 0.4$, a student calculates $(h - y)\vec{x}$ and then *adds* it to $\vec{\theta}$ as a gradient-ascent step. Explain the sign error. Compute the correct stochastic gradient-ascent update when $\vec{\theta} = (0, 0)^T$ and $\alpha = 0.1$.

Part II: Solutions

A. Linear Regression Solutions

L1.

$$h(4) = 1.5 + 0.8(4) = 1.5 + 3.2 = \boxed{4.7}.$$

L2.

$$h(2, 6) = -2 + 1.5(2) + 0.5(6) = -2 + 3 + 3 = \boxed{4}.$$

L3. The predictions are

$$2, \quad 3, \quad 4.$$

The residuals $h - y$ are

$$0, \quad 0.5, \quad 0.$$

Therefore

$$J = \frac{1}{2}\left(0^2 + 0.5^2 + 0^2\right) = \boxed{0.125}.$$

L4. The predictions are

$$2.2, \quad 2.9, \quad 3.6.$$

The residuals are

$$0.2, \quad 0.4, \quad -0.4.$$

Hence

$$J_B = \frac{1}{2}\left(0.2^2 + 0.4^2 + (-0.4)^2\right) = \frac{1}{2}(0.36) = \boxed{0.18}.$$

Since $0.125 < 0.18$, $\boxed{h_A \text{ has the smaller least-squares cost}}$.

L5.

$$h(2) = 1 + 0.5(2) = 2.$$

Thus

$$r = h - y = 2 - 5 = \boxed{-3}$$

and

$$J^{(i)} = \frac{1}{2}(-3)^2 = \boxed{4.5}.$$

L6. For a one-feature model,

$$\frac{\partial J^{(i)}}{\partial \theta_0} = h - y, \qquad \frac{\partial J^{(i)}}{\partial \theta_1} = (h - y)x.$$

Using $h - y = -3$ and $x = 2$,

$$\boxed{\frac{\partial J^{(i)}}{\partial \theta_0} = -3}, \qquad \boxed{\frac{\partial J^{(i)}}{\partial \theta_1} = -6}.$$

L7. Gradient descent uses

$$\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla J.$$

Therefore

$$\begin{bmatrix} 1 \\ 0.5 \end{bmatrix} - 0.1 \begin{bmatrix} -3 \\ -6 \end{bmatrix} = \boxed{\begin{bmatrix} 1.3 \\ 1.1 \end{bmatrix}}.$$

L8. At $(0,0)$, all predictions are 0. The three residuals are

$$0, \quad -1, \quad -2.$$

With input vectors $(1,-1)^T, (1,0)^T, (1,1)^T$,

$$\nabla J = 0\begin{bmatrix}1\\-1\end{bmatrix} - 1\begin{bmatrix}1\\0\end{bmatrix} - 2\begin{bmatrix}1\\1\end{bmatrix} = \boxed{\begin{bmatrix}-3\\-2\end{bmatrix}}.$$

L9.

$$\theta^{(1)} = \begin{bmatrix}0\\0\end{bmatrix} - 0.25\begin{bmatrix}-3\\-2\end{bmatrix} = \boxed{\begin{bmatrix}0.75\\0.50\end{bmatrix}}.$$

L10. At $(0.75, 0.50)$, the batch gradient is

$$\nabla J = \begin{bmatrix}3\theta_0 - 3\\2\theta_1 - 2\end{bmatrix} = \begin{bmatrix}-0.75\\-1\end{bmatrix}.$$

Thus

$$\theta^{(2)} = \begin{bmatrix}0.75\\0.50\end{bmatrix} - 0.25\begin{bmatrix}-0.75\\-1\end{bmatrix} = \boxed{\begin{bmatrix}0.9375\\0.75\end{bmatrix}}.$$

L11. At $(0.75, 0.50)$, the predictions are

$$0.25, \quad 0.75, \quad 1.25.$$

The residuals are

$$0.25, \quad -0.25, \quad -0.75.$$

Hence

$$J = \frac{1}{2}\left(0.25^2 + (-0.25)^2 + (-0.75)^2\right) = \boxed{0.34375}.$$

L12. Start with $\theta = (0,0)^T$.

For $(0,1)$:

$$h = 0, \quad h - y = -1,$$

so

$$\theta = \begin{bmatrix}0\\0\end{bmatrix} - 0.25\left[-1\begin{bmatrix}1\\0\end{bmatrix}\right] = \boxed{\begin{bmatrix}0.25\\0\end{bmatrix}}.$$

For $(1,2)$:

$$h = 0.25, \quad h - y = -1.75,$$

so

$$\theta = \begin{bmatrix}0.25\\0\end{bmatrix} - 0.25\begin{bmatrix}-1.75\\-1.75\end{bmatrix} = \boxed{\begin{bmatrix}0.6875\\0.4375\end{bmatrix}}.$$

For $(-1,0)$:

$$h = 0.6875 - 0.4375 = 0.25,$$

so the gradient is

$$0.25\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}0.25\\-0.25\end{bmatrix}.$$

Therefore

$$\theta = \begin{bmatrix}0.6875\\0.4375\end{bmatrix} - 0.25\begin{bmatrix}0.25\\-0.25\end{bmatrix} = \boxed{\begin{bmatrix}0.625\\0.500\end{bmatrix}}.$$

L13. **Batch gradient descent** forms a gradient using all training observations and then updates the parameters once. **Stochastic gradient descent** uses one observation and updates the parameters immediately after that observation.

L14. If α is too small, convergence can be very slow. If α is too large, the updates can overshoot, oscillate, or diverge.

L15.

$$\theta = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \boxed{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}.$$

No derivation of the formula is required.

L16.

$$h(2.5) = 1 + 2.5 = \boxed{3.5}.$$

The residual is

$$r = h - y = 3.5 - 3.8 = \boxed{-0.3}.$$

L17. With an intercept column, the design matrix and target vector are

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \vec{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad X^T = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Multiplication gives

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad X^T \vec{y} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The determinant of $X^T X$ is $3(5) - 3(3) = 6$, so

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}.$$

Consequently,

$$\vec{\theta} = (X^T X)^{-1} X^T \vec{y} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 25 - 18 \\ -15 + 18 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

The fitted line is $h_{\vec{\theta}}(x) = \frac{7}{6} + \frac{1}{2}x$. At $x = 3$, the prediction is $\frac{7}{6} + \frac{3}{2} = \boxed{\frac{8}{3}}$.

L18. Raw residuals can cancel: a residual of $+2$ and one of -2 sum to zero despite both being errors. Squaring makes every contribution nonnegative and penalizes larger errors more heavily.

L19. Training data are used to fit parameters. Validation data are used to choose model settings or compare candidate models. Test data provide a final estimate of performance on unseen examples. Repeated test-based selection leaks information from the test set into the modeling decisions and makes the reported test performance overly optimistic.

L20. The model may be overfitting the training observations or there may be a mismatch between the training and test distributions. Training error measures performance on data the model has already seen; test error measures generalization to held-out examples.

L21. θ_1 is the fitted slope: the change in predicted y for a one-unit increase in x. The prediction $h_{\vec{\theta}}(x) = \theta_0 + \theta_1 x$ depends on the particular input. A negative θ_1 means predictions decrease as x increases.

L22. Gradient descent does not enumerate candidate parameter vectors. It computes the gradient of the cost at the current parameters and moves in the opposite direction: $\vec{\theta} \leftarrow \vec{\theta} - \alpha \nabla J(\vec{\theta})$. Repeating the update can reduce the cost.

L23. Predictions at $x = 1, 2$ are 1 and 2. Residuals (prediction minus actual) are -1 and -2. Thus

$$J = \frac{1}{2}\big((-1)^2 + (-2)^2\big) = \frac{5}{2} = 2.5.$$

The conclusion is false. Zero cost requires every training residual to be zero; "small" nonzero residuals do not imply zero cost.

L24. No. An overly large learning rate can cause gradient descent to overshoot the minimum, making the cost increase even if the gradient is correct. Try a smaller α and examine whether the cost decreases. Increasing cost alone does not establish whether the derivative has been calculated correctly.

L25. All coordinates in a simultaneous gradient update must use the same old parameter vector. Initially $h = 0$, so $h - y = -5$. Therefore

$$\nabla J^{(i)} = (h - y)\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}-5\\-10\end{bmatrix}, \qquad \vec{\theta}_{\text{new}} = \begin{bmatrix}0\\0\end{bmatrix} - 0.1\begin{bmatrix}-5\\-10\end{bmatrix} = \begin{bmatrix}0.5\\1.0\end{bmatrix}.$$

Using an already updated coordinate while calculating another coordinate is not the simultaneous rule.

L26. Model A has the lower validation mean squared error ($2.4 < 4.8$), so it is preferred on the validation criterion. Model B's much lower training error but higher validation error suggests overfitting. The test set is reserved for final evaluation; repeated selection using its scores introduces test-set leakage and makes the evaluation optimistic. The table includes test results for interpretation, not as the basis for selecting the model.

L27. The claim reverses the direction of gradient descent. The update subtracts the gradient:

$$\theta_{1,\text{new}} = 1 - 0.2(3) = 0.4.$$

A positive partial derivative leads to a decrease in θ_1 for positive α.

B. Logistic Regression Solutions

G1. A linear model can produce any real number, including values below 0 or above 1. Such values cannot be interpreted directly as probabilities. Logistic regression keeps the linear score but passes it through the sigmoid, which always produces a value between 0 and 1.

G2.

$$z(3) = -1.2 + 0.8(3) = \boxed{1.2},$$

while

$$z(0) = -1.2 + 0.8(0) = \boxed{-1.2}.$$

G3.

$$g(1.2) = \frac{1}{1+e^{-1.2}} \approx \boxed{0.769},$$

$$g(-1.2) = \frac{1}{1+e^{1.2}} \approx \boxed{0.231}.$$

G4.

$$P(y=1 \mid \mathbf{x}) = \boxed{0.80}, \qquad P(y=0 \mid \mathbf{x}) = 1 - 0.80 = \boxed{0.20}.$$

Since $0.80 \geq 0.5$, the predicted class is $\boxed{1}$.

G5. The decision boundary occurs where $z = 0$:

$$-2 + 0.5x = 0.$$

Thus

$$0.5x = 2 \quad \Rightarrow \quad \boxed{x = 4}.$$

G6. The decision boundary is

$$\boxed{-3 + x_1 + 0.5x_2 = 0}.$$

At $(2, 4)$,

$$z = -3 + 2 + 0.5(4) = 1.$$

Therefore

$$P(y = 1 \mid \mathbf{x}) = g(1) \approx \boxed{0.731},$$

so the predicted class is $\boxed{1}$.

G7. For a true label of 1, use p; for a true label of 0, use $1 - p$. The observed-outcome probabilities are

$$0.90, \quad 0.80, \quad 0.70, \quad 0.60.$$

Thus

$$L = (0.90)(0.80)(0.70)(0.60) = \boxed{0.3024}.$$

G8. The observed-outcome probabilities are

$$0.60, \quad 0.60, \quad 0.55, \quad 0.55.$$

Hence

$$L = (0.60)(0.60)(0.55)(0.55) = \boxed{0.1089}.$$

Since $0.3024 > 0.1089$, $\boxed{\text{the predictor from G7 has the larger likelihood}}$.

G9. The observed-outcome probabilities are

$$1 - 0.20 = 0.80, \qquad 0.70, \qquad 0.90.$$

Therefore

$$L = (0.80)(0.70)(0.90) = \boxed{0.504}.$$

The log-likelihood is

$$\ell = \log(0.504) \approx \boxed{-0.685}.$$

G10. Write

$$g(z) = (1 + e^{-z})^{-1}.$$

Then

$$g'(z) = -(1 + e^{-z})^{-2}(-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}.$$

Also,

$$1 - g(z) = 1 - \frac{1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}}.$$

Therefore

$$g(z)(1 - g(z)) = \frac{1}{1 + e^{-z}} \frac{e^{-z}}{1 + e^{-z}} = \frac{e^{-z}}{(1 + e^{-z})^2} = \boxed{g'(z)}.$$

G11. For $y = 1$,

$$J = -\log h = -\log(0.80) \approx \boxed{0.223}.$$

G12. For $y = 0$,

$$J = -\log(1 - h) = -\log(0.20) \approx \boxed{1.609}.$$

G13.

$$J = -\left[0 \log(0.80) + (1 - 0) \log(1 - 0.80)\right] = -\log(0.20) \approx \boxed{1.609}.$$

This matches G12.

G14.

$$(y-h)\mathbf{x} = (1-0.60)\begin{bmatrix}1\\2\end{bmatrix} = 0.40\begin{bmatrix}1\\2\end{bmatrix} = \boxed{\begin{bmatrix}0.40\\0.80\end{bmatrix}}.$$

Gradient ascent gives

$$\theta_{\text{new}} = \theta_{\text{old}} + \alpha(y-h)\mathbf{x} = \begin{bmatrix}-0.5\\0.4\end{bmatrix} + 0.1\begin{bmatrix}0.4\\0.8\end{bmatrix} = \boxed{\begin{bmatrix}-0.46\\0.48\end{bmatrix}}.$$

G15. Initially $h^{(i)} = 0.5$ for all six observations. The intercept component of the average gradient is

$$\frac{1}{6}\left[(-0.5)+(-0.5)+(0.5)+(-0.5)+(0.5)+(0.5)\right] = 0.$$

The slope component is

$$\frac{1}{6}\left[(-0.5)(-2)+(-0.5)(-1)+(0.5)(0)+(-0.5)(1)+(0.5)(2)+(0.5)(3)\right].$$

The numerator is

$$1+0.5+0-0.5+1+1.5 = 3.5,$$

so

$$\frac{3.5}{6} = 0.5833.$$

Thus

$$\boxed{\nabla\ell_{\text{avg}} = \begin{bmatrix}0\\0.5833\end{bmatrix}}.$$

With $\alpha = 0.8$,

$$\theta^{(1)} = \begin{bmatrix}0\\0\end{bmatrix} + 0.8\begin{bmatrix}0\\0.5833\end{bmatrix} \approx \boxed{\begin{bmatrix}0\\0.4667\end{bmatrix}}.$$

G16. For $x = 2$,

$$z = 0 + 0.4667(2) = \boxed{0.9334}.$$

Hence

$$h = g(0.9334) \approx \boxed{0.718}.$$

Since $0.718 \geq 0.5$, the predicted class is $\boxed{1}$.

G17. The negative-log-likelihood gradient is

$$(h-y)\mathbf{x} = (0.70-0)\begin{bmatrix}1\\3\end{bmatrix} = \boxed{\begin{bmatrix}0.70\\2.10\end{bmatrix}}.$$

Gradient descent gives

$$\theta_{\text{new}} = \begin{bmatrix}0.2\\-0.1\end{bmatrix} - 0.05\begin{bmatrix}0.70\\2.10\end{bmatrix} = \boxed{\begin{bmatrix}0.165\\-0.205\end{bmatrix}}.$$

G18. If $J(\theta) = -\ell(\theta)$, then

$$\nabla J = -\nabla\ell.$$

Gradient descent on J is

$$\theta \leftarrow \theta - \alpha\nabla J = \theta - \alpha(-\nabla\ell) = \theta + \alpha\nabla\ell,$$

which is exactly gradient ascent on ℓ. Therefore the two procedures move the parameters in the same update direction.

G19. An ordinary linear model may output numbers below 0 or above 1, which cannot directly represent probabilities. Logistic regression first calculates a linear score $z = \vec{\theta}^T \vec{x}$ and maps it through $g(z) = 1/(1 + e^{-z})$, yielding a class-1 probability strictly between 0 and 1. A threshold such as 0.5 converts that probability into a class label of 0 or 1.

G20. The model assigns probability 0.74 to class 1 and probability $1 - 0.74 = 0.26$ to class 0. The value 0.74 is a probability, not a class; with a threshold of 0.5, the predicted class is 1.

G21. The sigmoid is monotonic and $g(0) = 0.5$. With a 0.5 threshold, the boundary is where $\vec{\theta}^T \vec{x} = 0$, or $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$, which is a line (assuming at least one nonzero feature coefficient).

G22. At threshold 0.5, both models correctly predict class 1 for the first case and class 0 for the second, so both have accuracy 100%. Their likelihoods differ because they assign different probabilities to the observed labels:

$$L_A = (0.90)(1 - 0.10) = 0.81, \qquad L_B = (0.55)(1 - 0.45) = 0.3025.$$

Classifier A assigns the greater likelihood.

G23. For a case with $y = 1$, the loss is $-\log h$. A mildly incorrect prediction of $h = 0.40$ gives loss $-\log(0.40) \approx 0.916$, whereas a confidently incorrect prediction of $h = 0.01$ gives loss $-\log(0.01) \approx 4.605$. The loss strongly penalizes giving a very low probability to the true outcome.

G24. A higher threshold requires a greater predicted class-1 probability before labeling a case as class 1. For example, $h = 0.60$ is labeled 1 at threshold 0.5 but 0 at threshold 0.7. The fitted probabilities remain unchanged; only the conversion from probability to class changes.

G25. The linear score z is not itself a probability. Apply the sigmoid:

$$P(y = 1 \mid \vec{x}) = g(-2) = \frac{1}{1 + e^2} \approx 0.119.$$

Since $0.119 < 0.5$, the predicted class is 0.

G26. Both predict class 1, so each is correct on this observation and has accuracy 100% on this one case. Their losses differ:

$$J_{0.51} = -\ln(0.51) \approx 0.673, \qquad J_{0.99} = -\ln(0.99) \approx 0.010.$$

The second model assigns far higher probability to the observed outcome and incurs lower log loss.

G27. Since the actual label is 0, the probability assigned to the observed outcome is

$$1 - h = 1 - 0.15 = 0.85.$$

Its negative-log-likelihood loss is $-\ln(0.85) \approx 0.163$. The output $h = 0.15$ is the probability of class 1, not of the observed class 0.

G28. The threshold probability 0.5 occurs when the *score* is zero, not necessarily when $x = 0$:

$$-1 + 0.5x = 0 \quad \Longrightarrow \quad x = 2.$$

Thus $x = 2$ is the decision boundary. Here $g(-1 + 0.5(2)) = g(0) = 0.5$.

G29. $(h - y)\vec{x}$ is the gradient of the *negative* log-likelihood for the observation. To maximize log-likelihood, use $(y - h)\vec{x}$, not $(h - y)\vec{x}$:

$$(y - h)\vec{x} = (1 - 0.4)\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}0.6\\1.2\end{bmatrix}.$$

Hence

$$\vec{\theta}_{\text{new}} = \vec{\theta} + \alpha(y - h)\vec{x} = \begin{bmatrix}0\\0\end{bmatrix} + 0.1\begin{bmatrix}0.6\\1.2\end{bmatrix} = \begin{bmatrix}0.06\\0.12\end{bmatrix}.$$